

Growth and Impact of Policy Making on Digital Payments (UPI) in India

A Multiple Linear Regression and Time-Series Study (2016–2026)

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Presentation Roadmap (20 Minutes)

1. Research Context & Objectives
2. Methodology: Data & Processing
3. Exploratory Data Analysis (EDA)
4. Hypothesis Testing
5. The Multiple Regression Model
6. Statistical Diagnostics & Refinement
7. Trends & Policy Insights
8. Conclusion

Introduction: The UPI Revolution

- **Context:** Unified Payments Interface (UPI) has transformed India's Digital Public Infrastructure (DPI).
- **The Research Question:** To what extent is this growth driven by organic adoption vs. specific policy interventions?
- **Timeline:** A high-resolution study covering the inception (April 2016) to projected trajectories (March 2026).

Theoretical Framework

Utilizing **Interrupted Time Series** logic within a **Multiple Linear Regression** framework to isolate the "bumps" caused by policy shifts.

Primary Sources:

- **RBI & NPCI:** Monthly statistics on volume and value.
- **PIB & Gazette of India:** Precise dates for policy implementations.

Dataset Structure:

- $N = 120$ (Monthly observations)
- Granularity: Monthly Time-Series
- Variables: Continuous Metrics + Binary Policy Dummies

To extract deeper insights, we derived the following from raw data:

① **Average Ticket Size (ATS):**

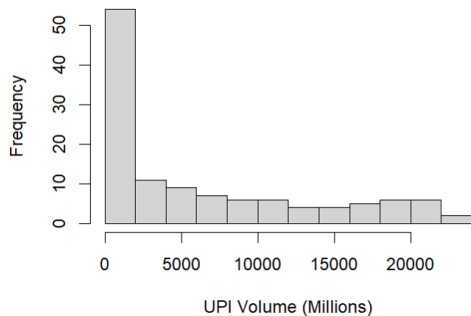
$$ATS = \frac{\text{Transaction Value (INR Crores)} \times 10^7}{\text{Transaction Volume (Millions)} \times 10^6} \quad (1)$$

② **Growth Velocity (MoM %):** Measuring percentage change in volume month-over-month.

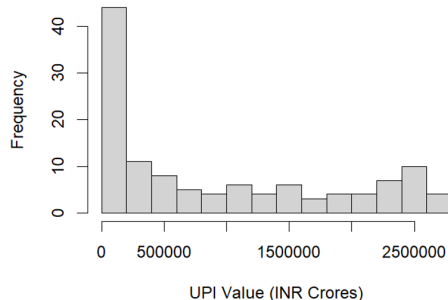
③ **Intervention Dummies:** Binary encoding (0, 1) for policies like Zero MDR, Credit Card on UPI, and COVID Lockdowns.

Distribution of Transaction Volume & Value

Histogram of UPI Transaction Volume

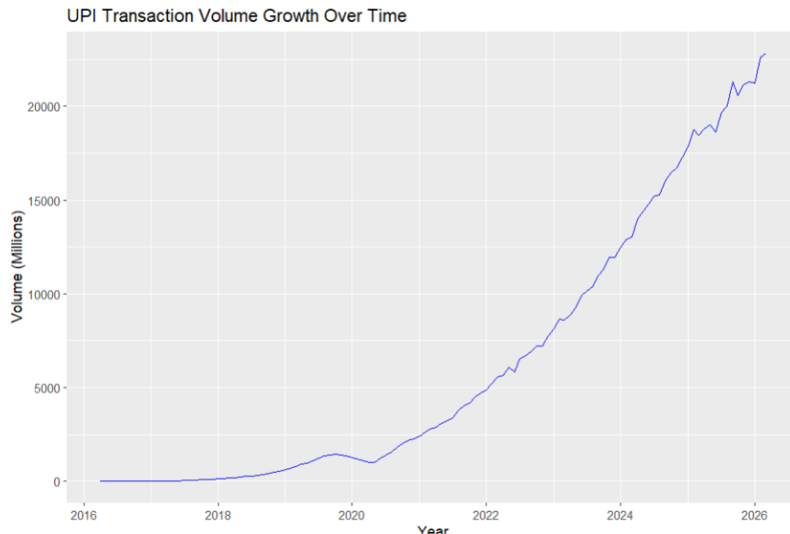


Histogram of UPI Transaction Value



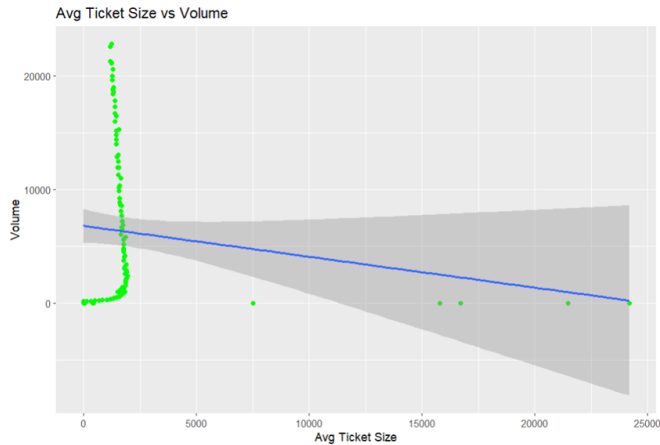
- **Observation:** Both Volume and Value exhibit extreme **Right Skewness**.
- **Statistical Implication:** The mean is significantly higher than the median, suggesting exponential growth rather than linear growth.

Visualizing the Structural Breaks



- **Pre-2020:** Slow, steady adoption.
- **Post-2020:** Steep vertical trajectory.
- **Questions for the Model:** Was this acceleration caused by COVID-19 or by the Zero

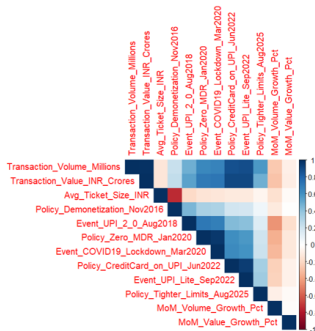
Average Ticket Size vs. Volume



The "Chai-Tapri" Effect:

- Scatter plot shows that as Volume increases, the Average Ticket Size does not necessarily increase at the same rate.
- This suggests UPI's scale is built on **high-frequency micro-payments**.

Correlation Heatmap: Early Diagnostics



Key Takeaways:

- **Volume vs. Value:** Correlation ≈ 1.0 . This is expected but indicates we must use only one as our dependent variable to avoid redundancy.
- **Policy Overlap:** High correlation between certain policy dates suggests potential **Multicollinearity** issues in the regression.

Since our growth data was not normally distributed, we used the **Wilcoxon Rank Sum Test**.

- H_0 : Growth rates are identical across pre- and post-2020 periods.
- H_1 : Growth rates are significantly different.
- **Result:** $p = 2.413 \times 10^{-6}$.

Decision

Reject H_0 . There is strong statistical evidence that the growth trajectory fundamentally shifted post-2020.

The initial Multiple Linear Regression equation:

$$Y_{Vol} = \beta_0 + \beta_1 X_{ATS} + \beta_2 D_{Demo} + \beta_3 D_{ZeroMDR} + \beta_4 D_{CC} + \beta_5 D_{Limits} + \beta_6 D_{COVID} + \epsilon$$

Where:

- Y_{Vol} : Transaction Volume (Millions)
- D : Dummy variables for policy interventions.
- ϵ : Error term.

Initial Model Summary

Variable	Estimate	t-value	p-value
Intercept	200.01	0.123	0.903
Avg Ticket Size	-0.0163	-0.154	0.878
Demonetization	270.11	0.167	0.868
Zero MDR	781.23	0.413	0.680
Credit Card on UPI	9443.53	14.40	0.001
Tighter Limits	8801.43	8.69	0.001
COVID Lockdown	1911.42	1.002	0.319

Model Metrics:

- $R^2 = 0.872$
- $F\text{-statistic} = 128.3$ ($p < 0.001$)

The Multicollinearity Problem

We suspected that Zero MDR and COVID-19 Lockdown variables were "fighting" for the same variance.

Variable	Variance Inflation Factor (VIF)
Zero MDR	14.84
COVID Lockdown	15.36
Credit Card on UPI	1.80
Avg Ticket Size	2.33

Threshold: $VIF > 10$ indicates severe multicollinearity. It makes our p -values unreliable for those specific variables.

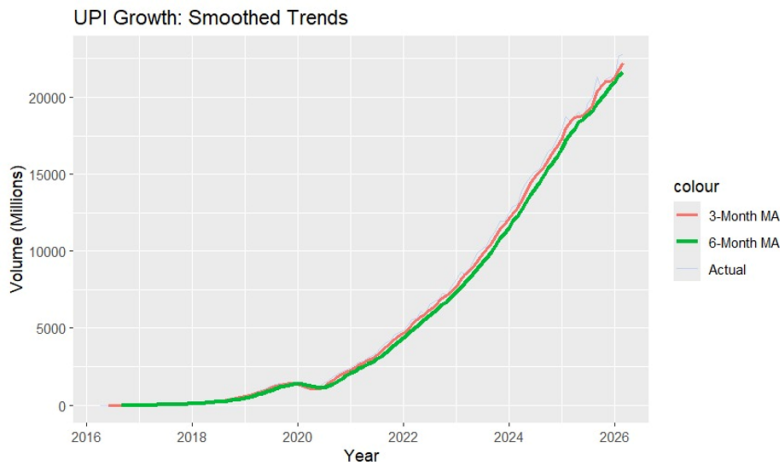
Decision

We removed the **Zero MDR** variable and re-ran the model using **COVID Lockdown** as a proxy for the structural shift of 2020.

Why?

- Reducing VIF increases the stability of the remaining coefficients.
- The R^2 remained high (0.87), indicating that removing the redundant variable did not cost us predictive power.
- Improved the reliability of the **t-tests** for individual policy impact.

Revealing the Underlying Trend: Smoothing



- **MA-3 & MA-6:** Moving averages reduce monthly volatility.
- **Insight:** Post-2020, the gap between the actual volume and the moving averages widened, proving the growth is not just a trend but an **acceleration**.

- ① **Policies that Matter:** Credit Card integration and Operational Limits are highly significant.
- ② **The Micro-Payment Proof:** Average Ticket Size is not a significant predictor ($p = 0.87$). UPI thrives because of **volume** (number of users), not because people are sending larger amounts.
- ③ **Early Interventions:** Demonetization acted as a catalyst, but its *independent* effect is overshadowed by later, more structured policies.

The Verdict: Influence or Correlation?

- **Bidirectional Influence:** Growth triggers policy, and policy accelerates growth.
- **Formalization:** The strong Volume-Value correlation indicates UPI is moving the informal cash economy into a **formal digital ledger**, aiding tax compliance and GDP tracking.
- **Financial Inclusion:** UPI is now a tool for **Credit Penetration** (Credit on UPI dummy significance).

Final Wrap-up:

- Our model successfully captures 87% of the growth dynamics.
- Statistical rigor was maintained by addressing multicollinearity.

Key References:

- Alonso et al. (2023) - IMF Working Paper (India Stack)
- NPCI Product Statistics (2016-2026)
- RBI Payments Vision 2025

Questions? Thank you for your time.